ABSTRACT

The **Procure-to-Pay (P2P) Procurement Management System** is designed to support **Minerva mobiles** and is designed to support Minerva Mobiles in efficiently managing the procurement of raw materials required for manufacturing new smartphone models. The system provides secure authentication for both employees and administrators, ensuring controlled and authorized access to procurement operations.

Employees can perform key procurement activities such as

- **Creating Requests for Quotation (RFQ).**
- **Generating Purchase Orders (PO).**
- **Processing Goods Receipt (GR).**
- **Viewing the Order Status (OS).**

Administrators are responsible for

- **Monitoring inventory or stock reports.**

By integrating and streamlining procurement operations into single system, the solution improves

- **Process Transparency**
- **Operational Efficiency**
- **Data accuracy**

The system minimizes manual intervention, enhances coordination between procurement and inventory management and supports timely decision-making, thereby strengthening the overall effectiveness of the Procure-to-Pay process.

INTRODUCTION

Procurement is an important part of any manufacturing process, as it ensures that the required materials are available when needed. For Minerva Mobiles, which is planning to launch a new smartphone model, managing the purchase of raw materials in an organized way is essential to keep production running without interruptions. A smooth procurement process helps the company save time, reduce errors, and maintain better coordination between teams.

As the number of procurement activities increases, handling them manually can become confusing and inefficient. Using a digital procurement system makes these tasks easier by bringing everything into one place and providing clear visibility of orders and stock. This project focuses on developing such a system to help employees manage procurement activities efficiently and assist administrators in monitoring inventory, leading to smoother operations and better overall management.

OBJECTIVES OF PROJECT

- To design and implement a Procure-to-pay (P2P) procurement management system in the SAP S/4 HANA environment.
- To develop application logic using ABAP programming for RFQ creation, Purchase Order processing, and Goods Receipt posting.
- To utilize SAP HANA in-memory capabilities for faster data processing and reporting.
- To create CDS views for efficient data modeling, real-time analytics, and stock visibility.
- To implement role-based authorization for admin and employee access in SAP.
- To enable admin users to monitor plant storage location stock using HANA-optimized ABAP artifacts.
- To increase system performance using CDS-based push-down techniques instead of classical ABAP loops.
- To build or integrate Fiori/SAP GUI screens for user-friendly procurement transactions.
- To automate procurement steps to reduce manual effort and processing time.
- To ensure data consistency and transparency across RFQ, PO, GR and stock management processes.

OUR CONTRIBUTION

- Designed the overall workflow of the Procure-to-Pay process covering RFQ creation, purchase order generation, goods receipt, and order status tracking.

- Developed a role-based login system for employees and administrators to ensure secure and authorized access.

- Implemented employee functionalities such as creating RFQs, generating Pos, performing Goods Receipt, updating order status, and viewing order
  Details.

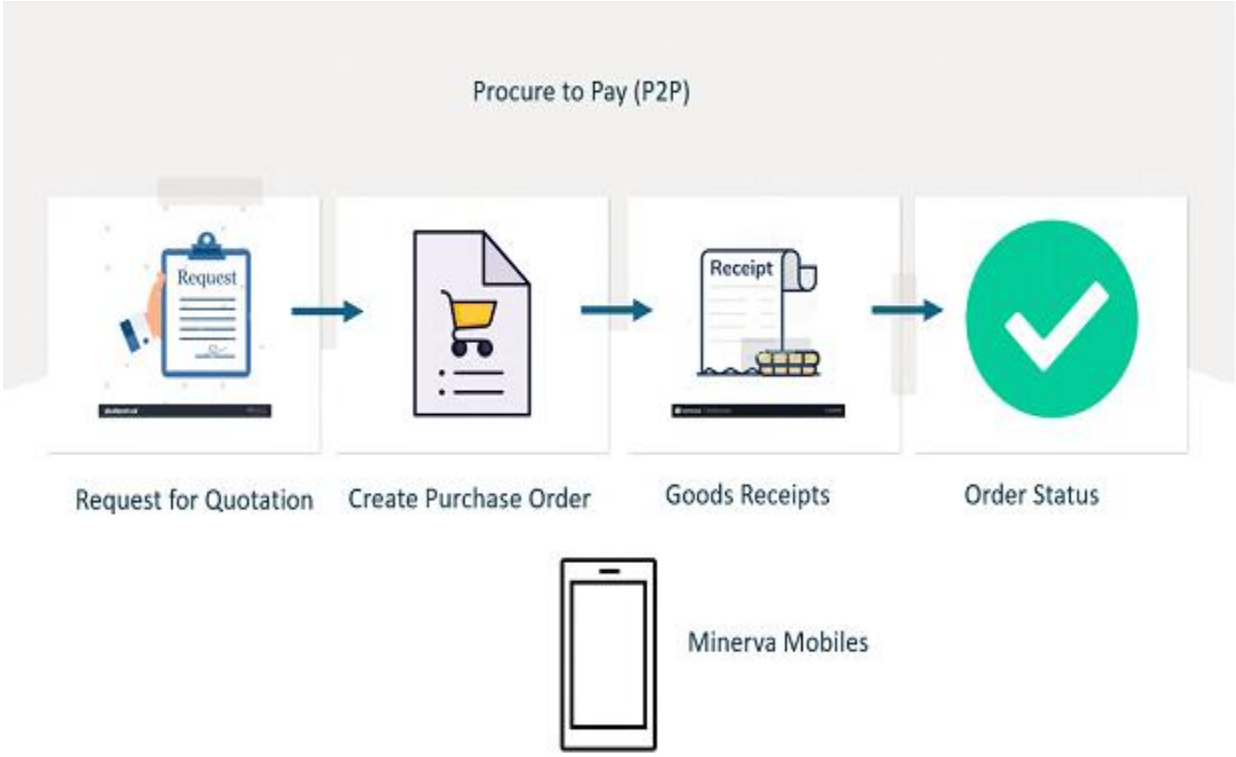
- Implemented admin functionality to monitor available stock in the plant storage location for effective inventory processes.

- Created a user-friendly interface to simplify procurement activities and reduce manual effort.

- Integrated all modules to ensure smooth data flow and real-time updates across procurement and inventory processes.

- Tested the system to verify functionality, accuracy, and reliability of procurement and inventory processes.

PROJECT FLOW



SYSTEM OVERVIEW

- The system supports the Procure-to-Pay (P2P) cycle for purchasing raw materials required for smartphone manufacturing.
- It provides a role-based login where both admin and employee can securely access the application.
- Employees can create requests for Quotation (RFQ) to collect price and material information from vendors.
- Employees can generate Purchase Orders (PO) based on approved quotations.
- The system allows employees to perform Goods Receipt (GR) after materials are delivered.
- Employees can update order status and view the current status of all orders placed.
- The admin has access to view and monitor available stock in the plant storage location.
- The system aims to automate procurement activities, reduce manual effort, and minimize errors.
- It provides real-time visibility of orders, receipts, and stock levels.
- The system supports improved decision-making and efficient inventory management.

TECHNOLOGIES USED

The Procure-to-Pay project is implemented using modern SAP technologies to ensure efficient data modeling, real-time access, and seamless integration of procurement processes.

1. **SAP S/4HANA**

   SAP S/4HANA is the core ERP platform used in this project. It provides a simplified data model and high-performance in-memory processing, enabling real-time handling of procurement transactions such as purchase requisitions, purchase orders, and goods receipt.

2. **ABAP Core Data Services (CDS Views)**

   CDS Views are used as the primary data modeling layer in this project. They define and structure procurement-related business data in a reusable and efficient manner. CDS views help in:
   - Modeling procurement entities
   - Improving performance through database-level processing
   - Enabling real-time data access
   -

3. **OData Services**

   OData services are created by exposing CDS views. They provide a standardized, REST-based interface to access procurement data. OData services enable:
   - Real-time data consumption
   - Integration with user interfaces
   - Interactive and dynamic data retrieval

4. **ABAP Development Tools (ADT) in Eclipse**

   Eclipse with ADT is used to develop CDS views and manage OData services. It provides a modern development environment with features such as syntax highlighting, data preview, and error checking.

5. **SAP Fiori / ALV Output**

   SAP Fiori or ALV-based output screens are used to display procurement data in an interactive format. These interfaces support filtering, sorting, and drill-down capabilities for better data analysis.

6. **SAP HANA Database**

   SAP HANA is the underlying in-memory database used by SAP S/4HANA. It supports high-speed data processing and allows complex operations to be executed efficiently at the database level.

Database Table Creation

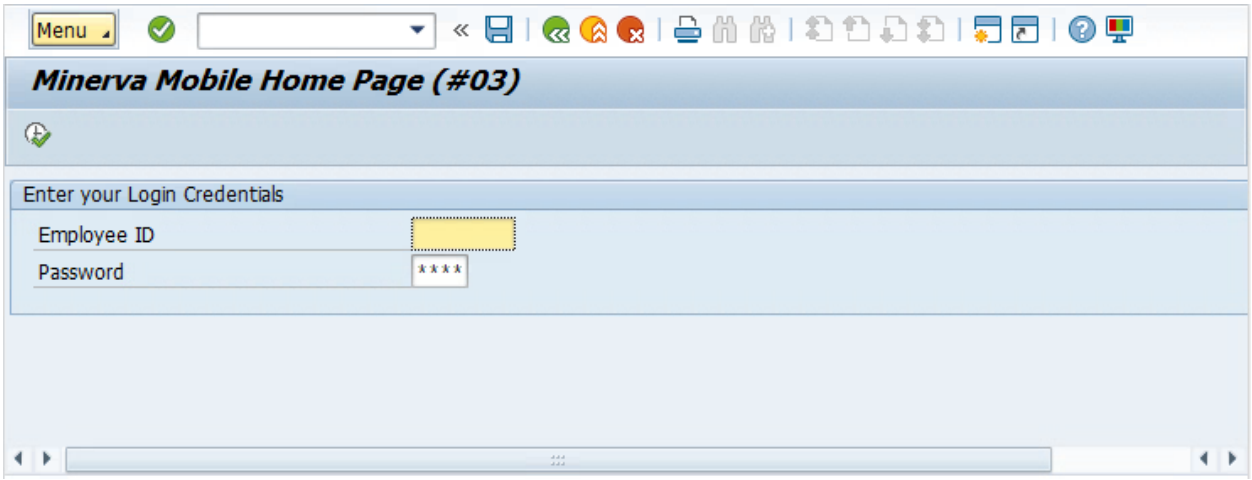
The Project begins with the creation and identification of required database tables to store procurement-related master and transactional data for Minerva Mobiles. The tables that are created are

- **Employee table**
- **Plant table**
- **Storage Location table**
- **Material table**
- **Vendor table**
- **Stock details table**
- **Request For Quotation (RFQ) table**
- **Purchase order table**
- **Goods receipt table**

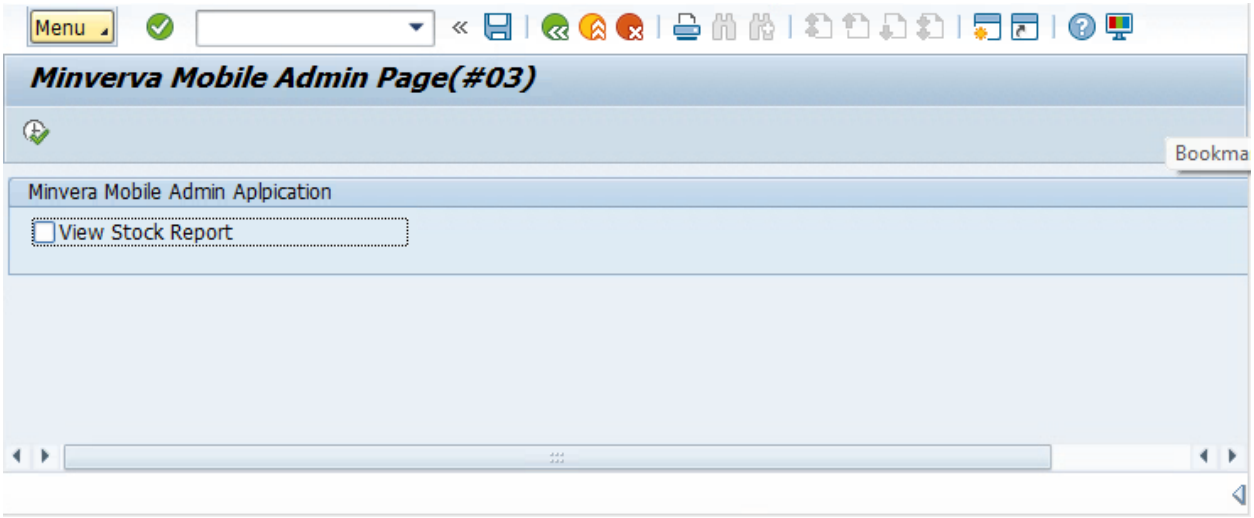
The data is entered into the tables by using **Table Maintenance Generator (TMG).**

OUTPUTS

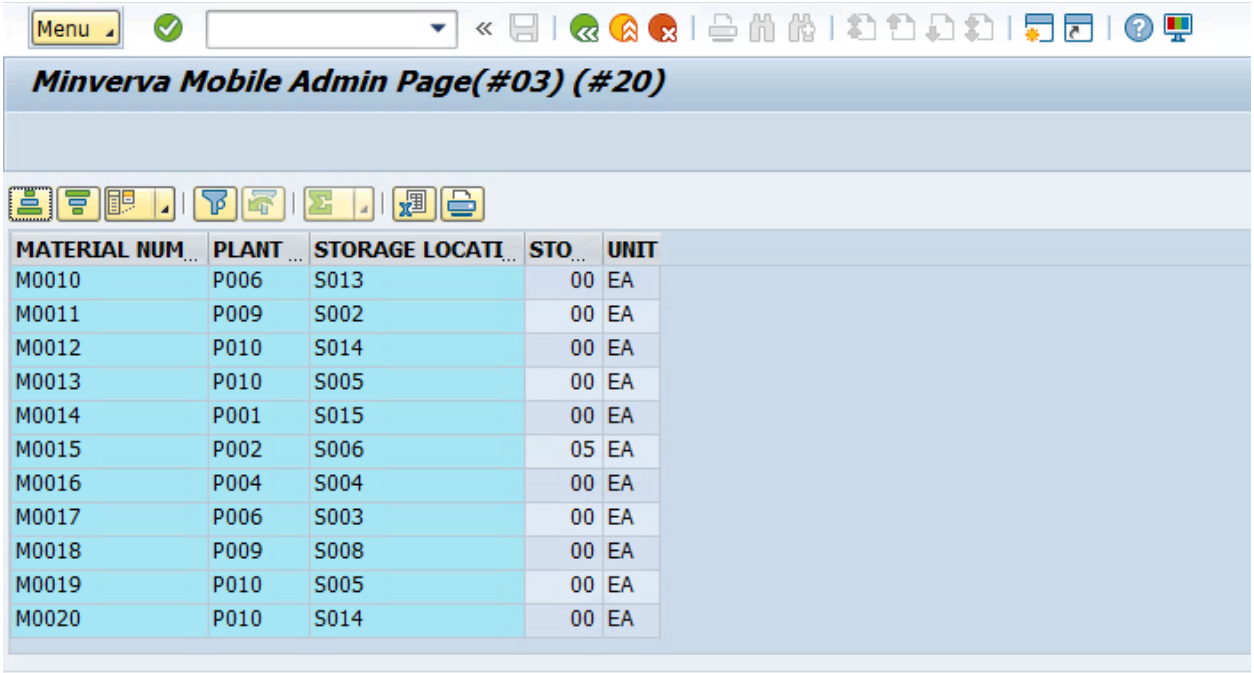
Home Screen



Admin Screen

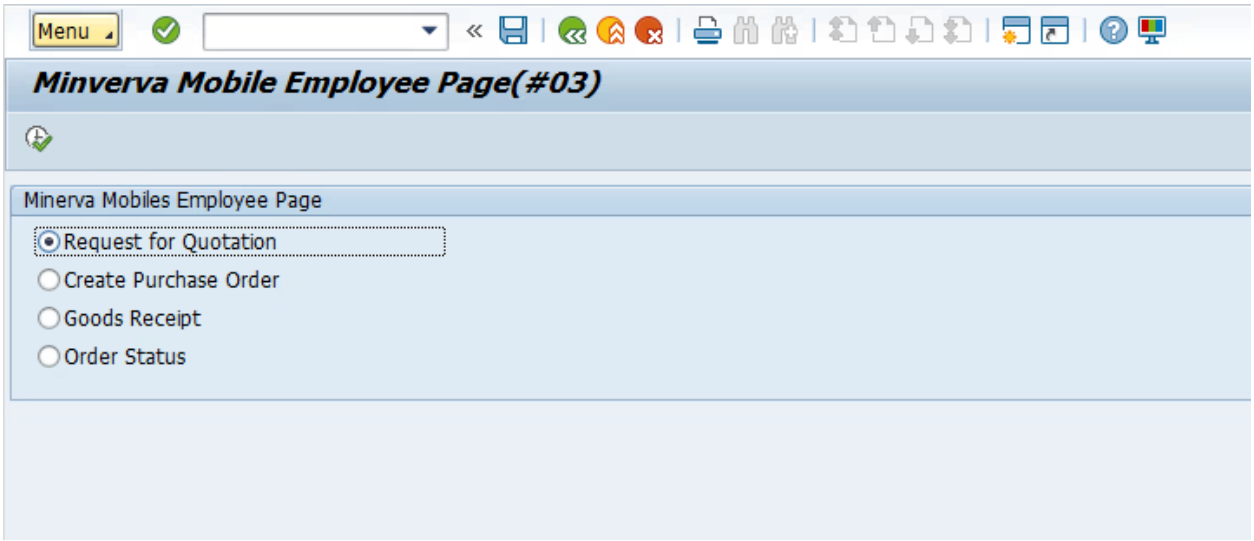


Admin Screen Output details:



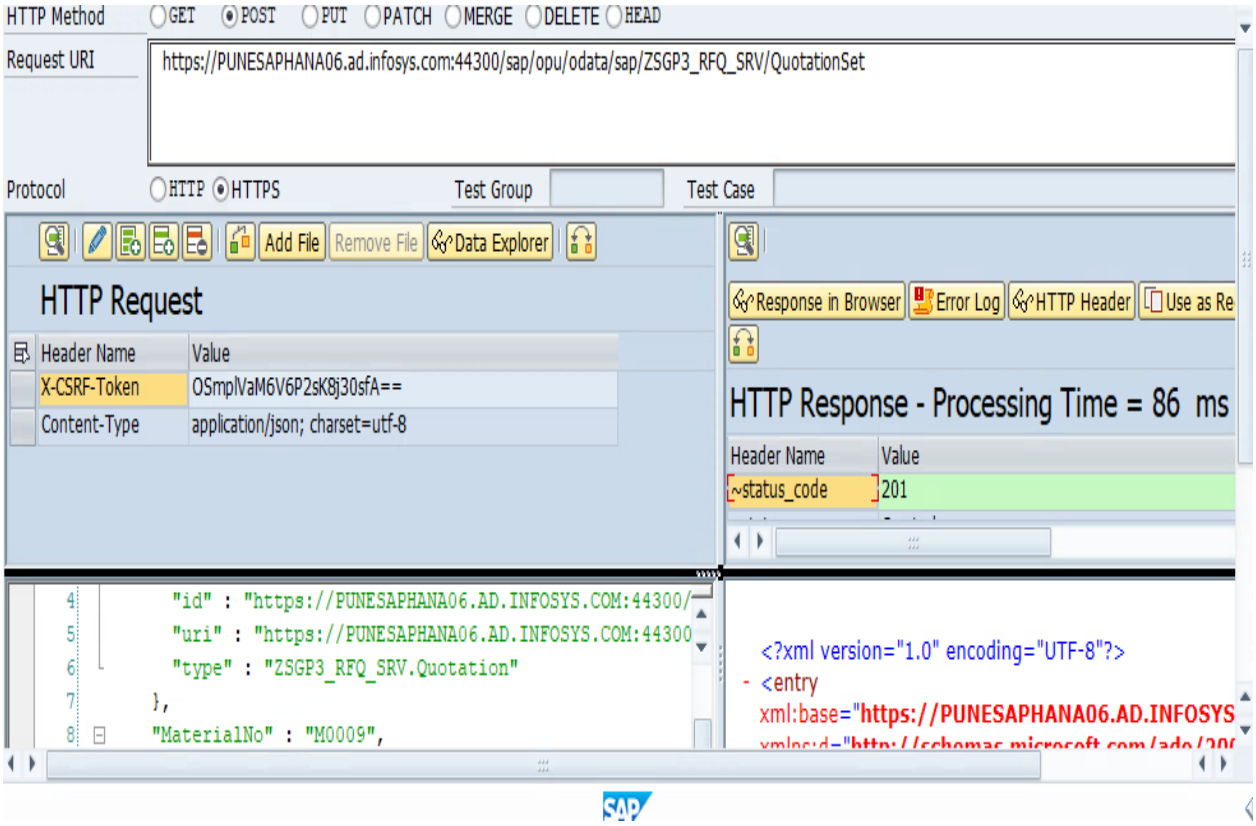
MATERIAL NUM...	PLANT ...	STORAGE LOCATI...	STO...	UNIT
M0010	P006	S013	00	EA
M0011	P009	S002	00	EA
M0012	P010	S014	00	EA
M0013	P010	S005	00	EA
M0014	P001	S015	00	EA
M0015	P002	S006	05	EA
M0016	P004	S004	00	EA
M0017	P006	S003	00	EA
M0018	P009	S008	00	EA
M0019	P010	S005	00	EA
M0020	P010	S014	00	EA

Employee Screen with four Radio Buttons



RFQ option's output: Creating the RFQ table using ODATA

CREATE_ENTITY for RFQ (Request for Quotation), Automatic *RFQNO generation*



GET_ENTITY for RFQ (Request for Quotation)

SAP Gateway Client

Execute | Select | Service Administration | Service Implementation | Switch User | Entity Set | Add URI Option

HTTP Method: GET POST PUT PATCH MERGE DELETE HEAD

Request URI: /sap/opu/odata/sap/ZSGP3_RFQ_SRV/QuotationSet(RfqNo='5000000009',MaterialNo='M0008')

Protocol: HTTP HTTPS Test Group Test Case

Add File | Remove File | Data Explorer

HTTP Request

Response in Browser | Error Log | HTTP Header | Use as Request | Data Explorer

HTTP Response - Processing Time = 84 ms

Header Name	Value
~status_code	200
~status_reason	OK

```
- <content type="application/xml">
  - <m:properties>
      <d:RfqNo>5000000009</d:RfqNo>
      <d:MaterialNo>M0008</d:MaterialNo>
      <d:RfqDate>2026-01-12T00:00:00</d:RfqDate>
      <d:VendorId>V0004</d:VendorId>
      <d:ValidityStart>2026-01-01T00:00:00</d:ValidityStart>
      <d:ValidityEnd>2026-12-31T00:00:00</d:ValidityEnd>
      <d:Quantity>100</d:Quantity>
```

UPDATE_ENTITY for RFQ (Request for Quotation)

HTTP Method: GET POST PUT PATCH MERGE DELETE HEAD

Request URI: https://PUNESAPHANA06.ad.infosys.com:44300/sap/opu/odata/sap/ZSGP3_RFQ_SRV/QuotationSet(RfqNo='5000000009',MaterialNo='M0008')

Protocol: HTTP HTTPS Test Group Test Case

Add File | Remove File | Data Explorer

HTTP Request

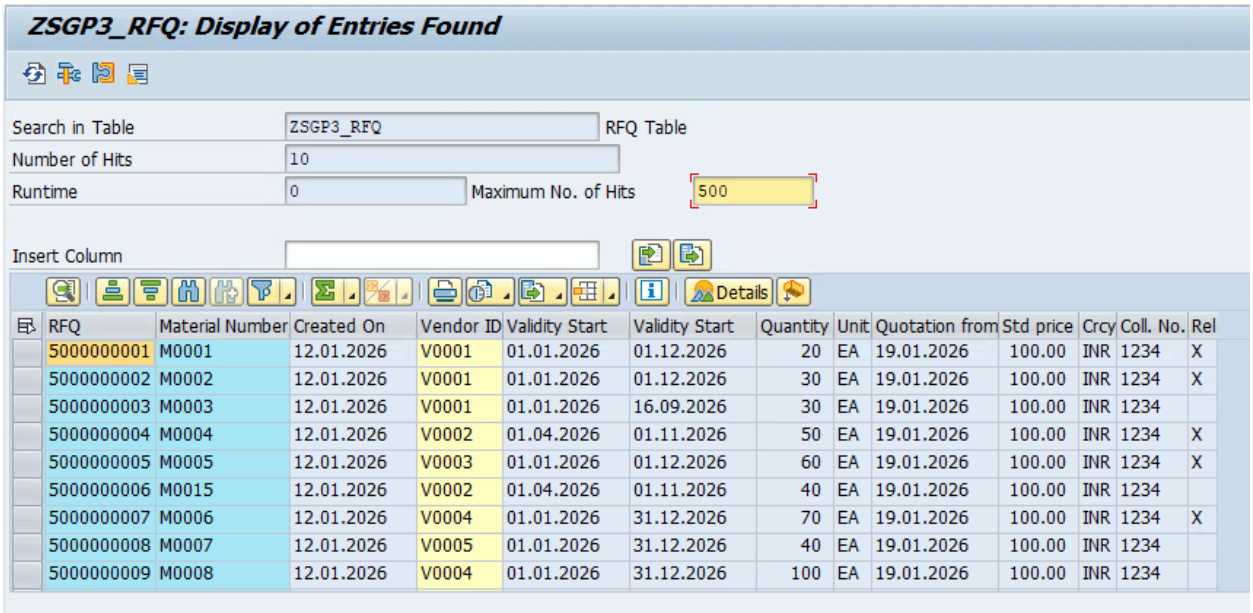
Response in Browser | Error Log | HTTP Header | Use as Request | Data Explorer

HTTP Response - Processing Time = 83 ms

Header Name	Value
~status_code	204
~status_reason	No Content

```
1 <?xml version="1.0" encoding="utf-8
2 <entry xml:base="http://PUNESAPHANA
3  <id>http://PUNESAPHANA06.ad.infosy
4  <title type="text">QuotationSet(Rf
5  <updated>2026-01-15T16:36:21Z</upd
6  <category term="ZSGP3_RFQ_SRV.Quot
7  <link href="QuotationSet(RfqNo='50
8  <content type="application/xml">
```

Total Entries Display (Tcode : SE16N)

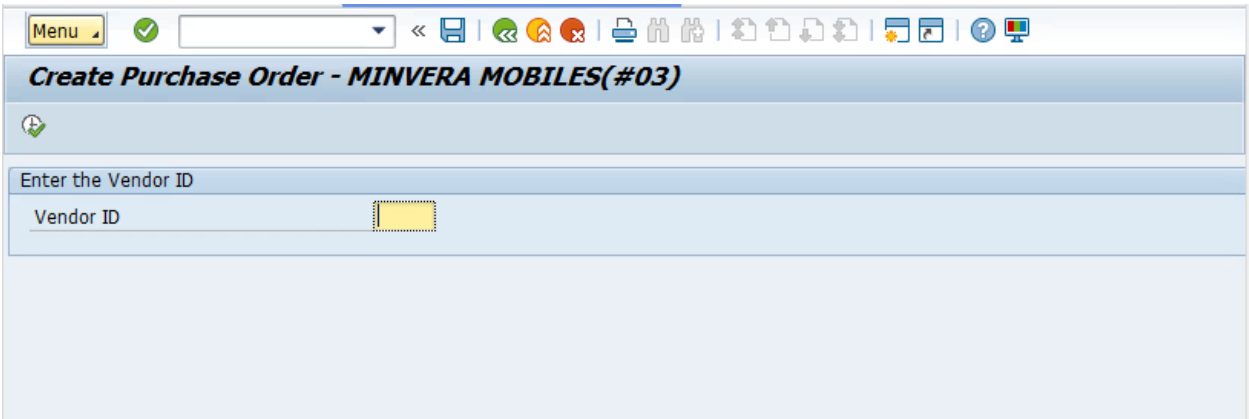


ZSGP3_RFQ: Display of Entries Found

Search in Table: ZSGP3_RFQ — RFQ Table
Number of Hits: 10
Runtime: 0 — Maximum No. of Hits: 500
Insert Column

RFQ	Material Number	Created On	Vendor ID	Validity Start	Validity Start	Quantity	Unit	Quotation from	Std price	Crcy	Coll. No.	Rel
5000000001	M0001	12.01.2026	V0001	01.01.2026	01.12.2026	20	EA	19.01.2026	100.00	INR	1234	X
5000000002	M0002	12.01.2026	V0001	01.01.2026	01.12.2026	30	EA	19.01.2026	100.00	INR	1234	X
5000000003	M0003	12.01.2026	V0001	01.01.2026	16.09.2026	30	EA	19.01.2026	100.00	INR	1234	
5000000004	M0004	12.01.2026	V0002	01.04.2026	01.11.2026	50	EA	19.01.2026	100.00	INR	1234	X
5000000005	M0005	12.01.2026	V0003	01.01.2026	01.12.2026	60	EA	19.01.2026	100.00	INR	1234	X
5000000006	M0015	12.01.2026	V0002	01.04.2026	01.11.2026	40	EA	19.01.2026	100.00	INR	1234	
5000000007	M0006	12.01.2026	V0004	01.01.2026	31.12.2026	70	EA	19.01.2026	100.00	INR	1234	X
5000000008	M0007	12.01.2026	V0005	01.01.2026	31.12.2026	40	EA	19.01.2026	100.00	INR	1234	
5000000009	M0008	12.01.2026	V0004	01.01.2026	31.12.2026	100	EA	19.01.2026	100.00	INR	1234	

Purchase Order option's output: Purchase Order details as per given vendor ID

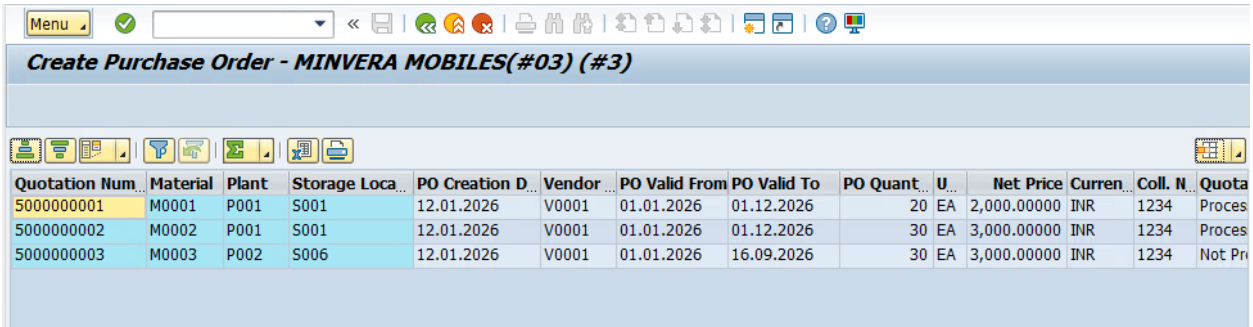


Menu

Create Purchase Order - MINVERA MOBILES(#03)

Enter the Vendor ID

Vendor ID

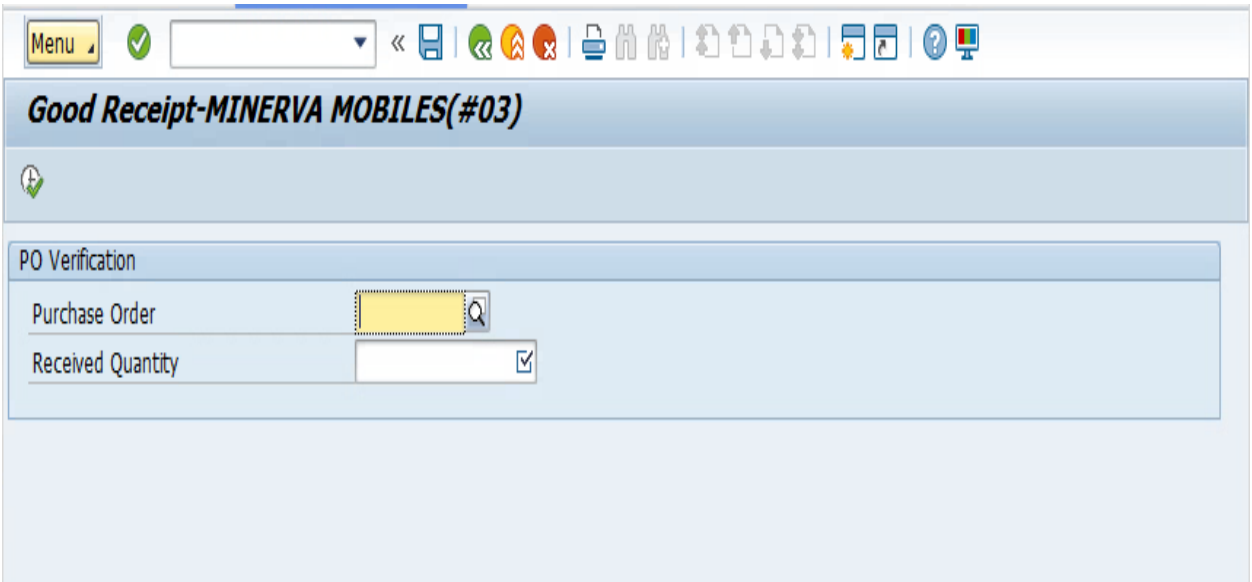


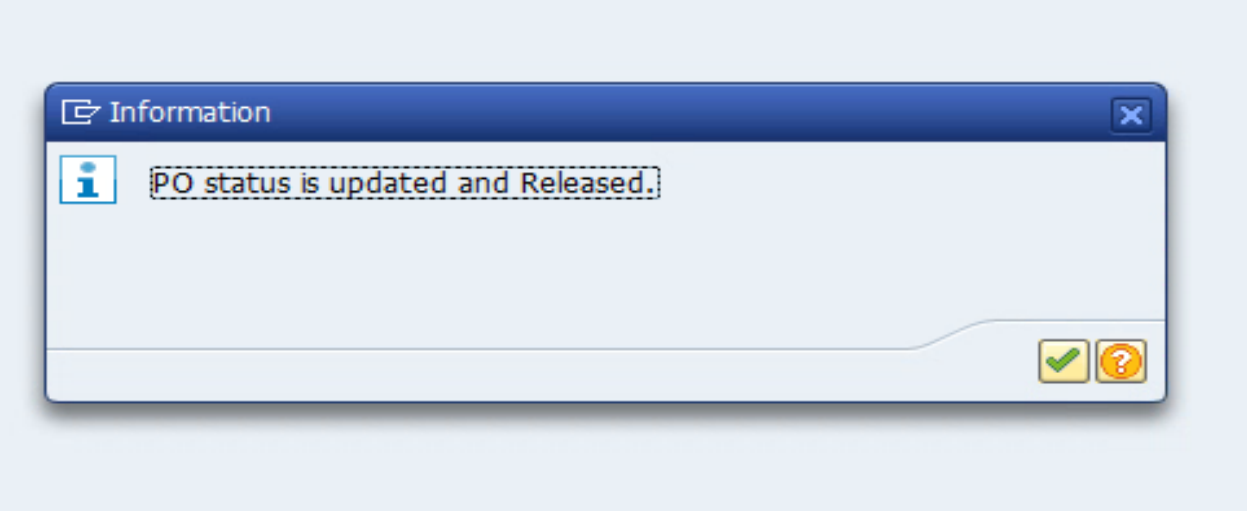
Menu

Create Purchase Order - MINVERA MOBILES(#03) (#3)

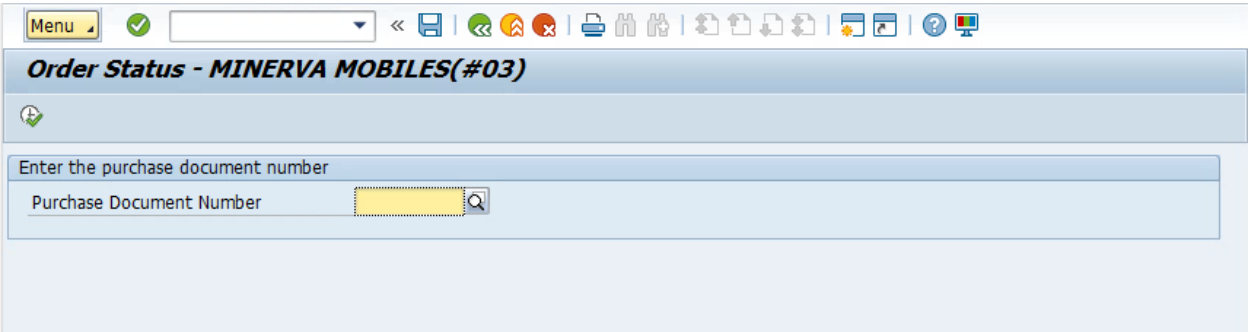
Quotation Num	Material	Plant	Storage Loca	PO Creation D	Vendor	PO Valid From	PO Valid To	PO Quant	U	Net Price	Curren	Coll. N	Quota
5000000001	M0001	P001	S001	12.01.2026	V0001	01.01.2026	01.12.2026	20	EA	2,000.00000	INR	1234	Proces
5000000002	M0002	P001	S001	12.01.2026	V0001	01.01.2026	01.12.2026	30	EA	3,000.00000	INR	1234	Proces
5000000003	M0003	P002	S006	12.01.2026	V0001	01.01.2026	16.09.2026	30	EA	3,000.00000	INR	1234	Not Pr

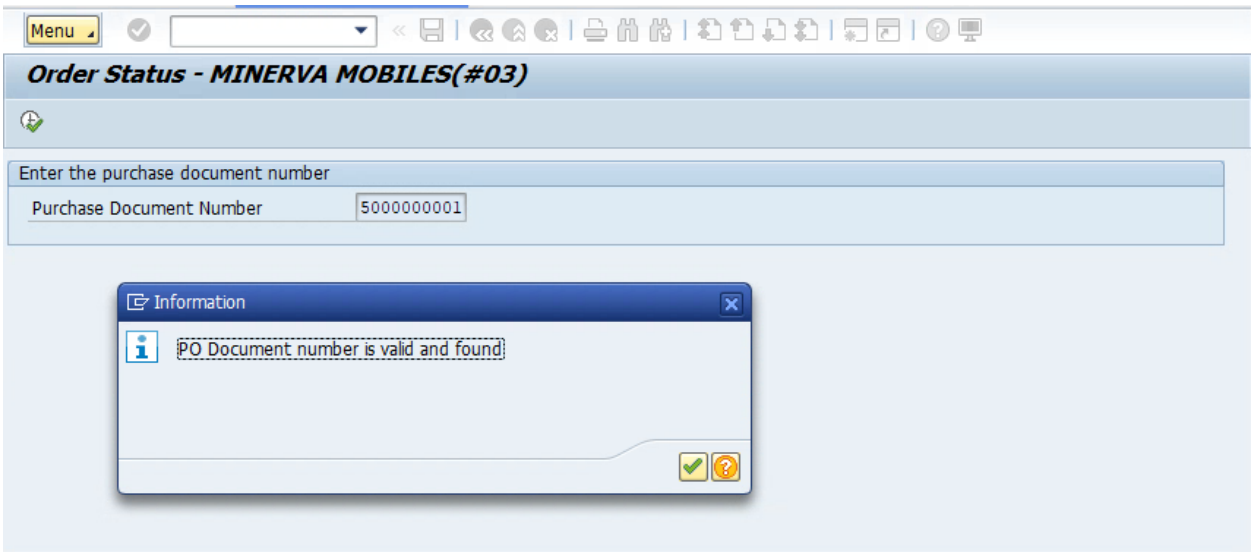
Goods receipt option's output: Giving the required quantity for a particular purchase order id.

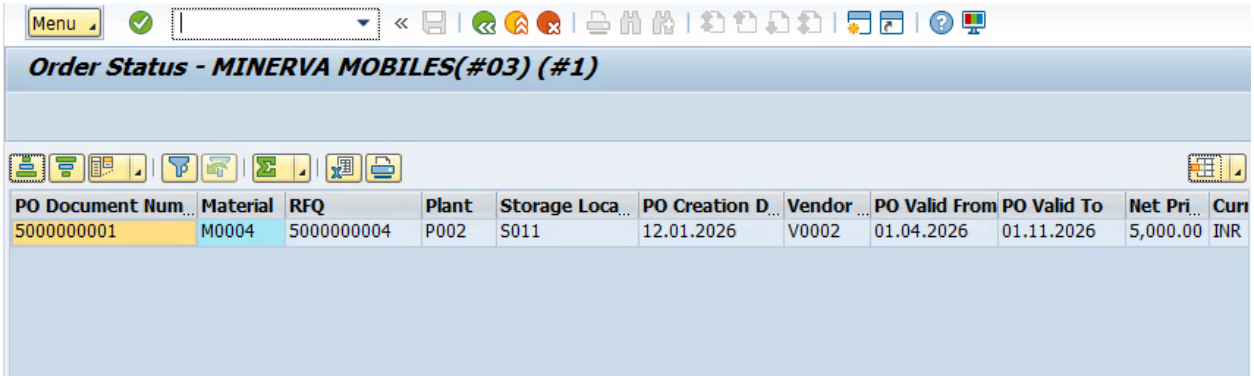




Order status option's output: Displaying the order status for given purchase document number

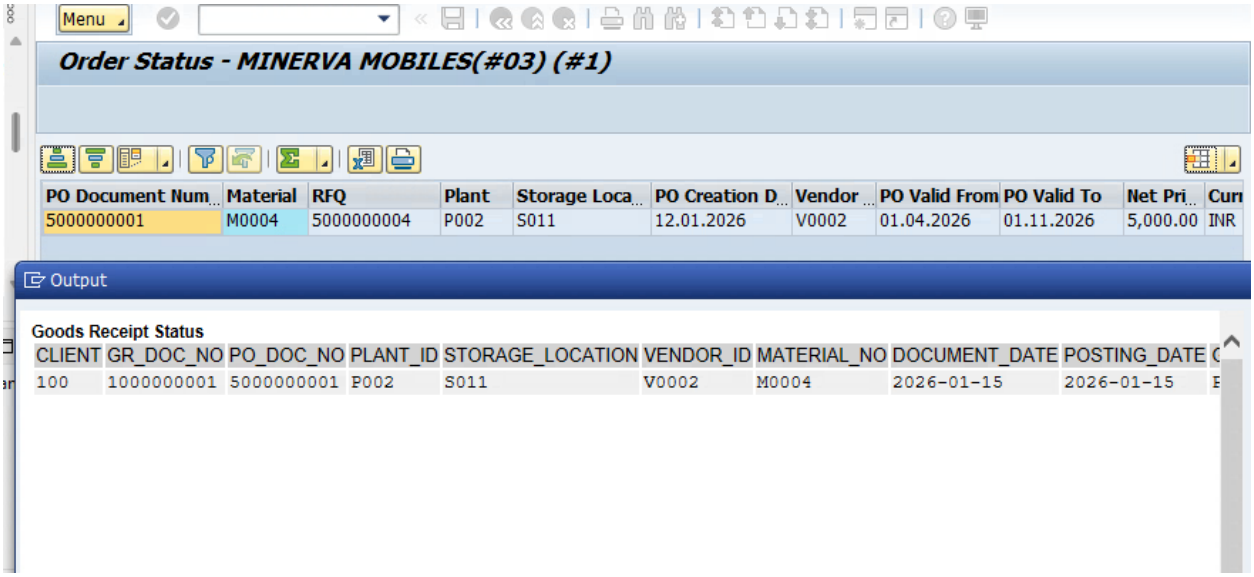






PO Document Num...	Material	RFQ	Plant	Storage Loca...	PO Creation D...	Vendor ...	PO Valid From	PO Valid To	Net Pri...	Curr
5000000001	M0004	5000000004	P002	S011	12.01.2026	V0002	01.04.2026	01.11.2026	5,000.00	INR

Displaying the records after double clicking on the PO document number from the Goods receipt table.



PO Document Num...	Material	RFQ	Plant	Storage Loca...	PO Creation D...	Vendor ...	PO Valid From	PO Valid To	Net Pri...	Curr
5000000001	M0004	5000000004	P002	S011	12.01.2026	V0002	01.04.2026	01.11.2026	5,000.00	INR

Goods Receipt Status

CLIENT	GR_DOC_NO	PO_DOC_NO	PLANT_ID	STORAGE_LOCATION	VENDOR_ID	MATERIAL_NO	DOCUMENT_DATE	POSTING_DATE	C
100	1000000001	5000000001	P002	S011	V0002	M0004	2026-01-15	2026-01-15	E

ODATA for Goods Receipt which includes navigation for materials and stock table.

GET_ENTITY METHOD:

Execute | Select | Service Administration | Service Implementation | Switch User | Entity Set | Add URI Option

HTTP Method: GET POST PUT PATCH MERGE DELETE HEAD

Request URI: /sap/opu/odata/sap/ZSGP3_VIEWSTOCKS_SRV/MaterialDetailsSet('M0001')

Protocol: HTTP HTTPS Test Group Test Case

Add File | Remove File | Data Explorer

HTTP Request

Response in Browser | Error Log | HTTP Header | Use as Request | Data Explorer

HTTP Response - Processing Time = 63 ms

Header Name	Value
~status_code	200
~status_reason	OK

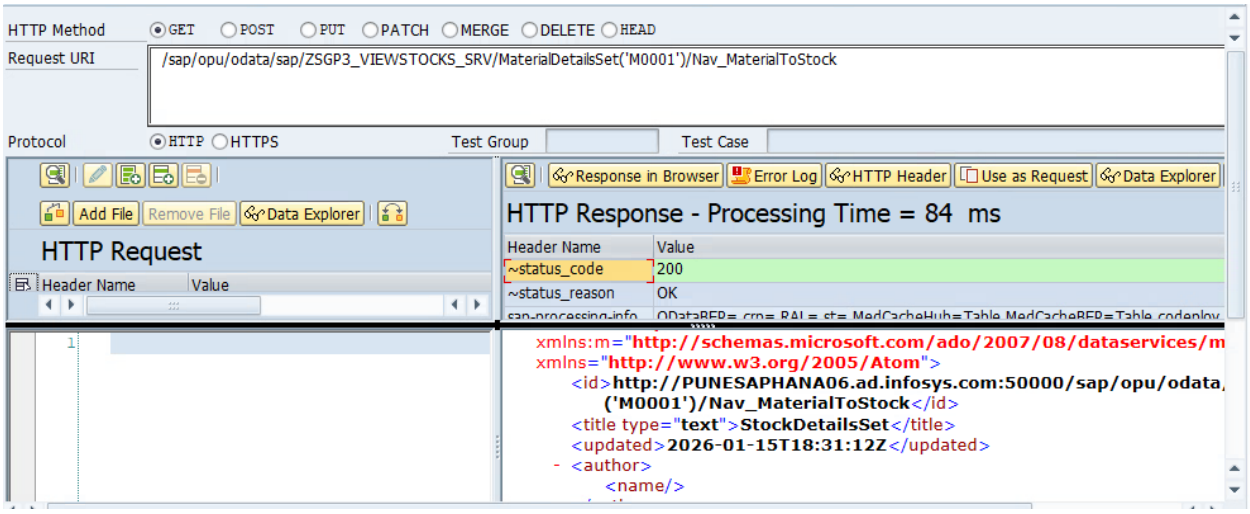
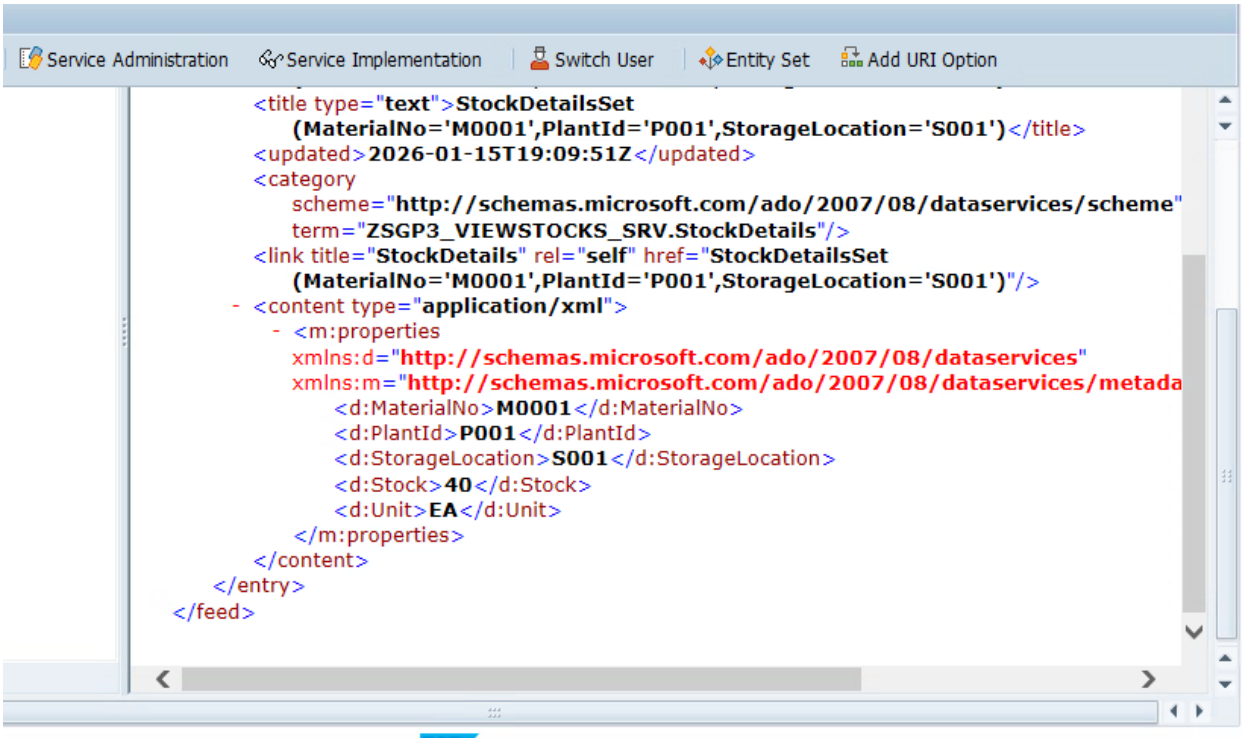

```
- <entry
  xml:base="http://PUNESAPHANA06.ad.infosys.com:50000/sap/opu/oda
  xmlns:d="http://schemas.microsoft.com/ado/2007/08/dataservices"
  xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/m
  xmlns="http://www.w3.org/2005/Atom">
    <id>http://PUNESAPHANA06.ad.infosys.com:50000/sap/opu/odata/
      ('M0001')</id>
    <title type="text">MaterialDetailsSet('M0001')</title>
    <updated>2026-01-15T17:27:13Z</updated>
```


SAP Gateway Client

Execute | Select | Service Administration | Service Implementation | Switch User | Entity Set | Add URI Option


```
  <id>http://PUNESAPHANA06.ad.infosys.com:50000/sap/opu/odata/sap/ZSGP3_VIE
    ('M0001')</id>
  <title type="text">MaterialDetailsSet('M0001')</title>
  <updated>2026-01-15T19:08:35Z</updated>
  <category
    scheme="http://schemas.microsoft.com/ado/2007/08/dataservices/scheme"
    term="ZSGP3_VIEWSTOCKS_SRV.MaterialDetails"/>
  <link title="MaterialDetails" rel="self" href="MaterialDetailsSet('M0001')"/>
  <link title="Nav_MaterialToStock" type="application/atom+xml;type=feed"
    rel="http://schemas.microsoft.com/ado/2007/08/dataservices/related/Nav_Ma
    href="MaterialDetailsSet('M0001')/Nav_MaterialToStock"/>
- <content type="application/xml">
  - <m:properties>
      <d:MaterialNo>M0001</d:MaterialNo>
      <d:MaterialName>Bluetooth Module</d:MaterialName>
      <d:MaterialType>ELEC</d:MaterialType>
      <d:MaterialGroup>E</d:MaterialGroup>
      <d:CreatedNo>2024-01-01T00:00:00</d:CreatedNo>
    </m:properties>
  </content>
</entry>
```


1 Col 1

Displaying Data using Navigation

Nav_MaterialToStock: